

DS 6021: Introduction to Predictive Modeling

Lab 0: From Messy Data to Modeling-Ready Data

Fall 2026

Class Data

For this lab, you will work with a dataset generated by **your DS 6021 class**.

The data come from the class survey completed at the beginning of the semester. Because the responses were entered by real people rather than generated from a perfectly curated dataset, you should expect inconsistencies, missing values, unexpected entries, and variables stored in inconvenient formats.

That is exactly the point.

In practice, data rarely arrive ready for modeling. Before fitting a predictive model, a data scientist must first understand where the data came from, clean and organize them, explore their distributions, and identify potentially useful relationships.

Class Survey: [DS 6021 Class Survey](#)

Learning Outcomes

By the end of this lab, you should be able to:

1. **Import and inspect a real-world dataset** using Python.
2. **Identify data-quality issues** such as inconsistent entries, inappropriate data types, and missing values.
3. **Clean and transform variables** into forms appropriate for analysis.
4. **Summarize numerical and categorical variables** using appropriate descriptive statistics.
5. **Create appropriate visualizations** for numerical and categorical data.
6. **Explore relationships among variables** using bivariate and multivariate visualizations.
7. **Distinguish potential predictor variables from outcome variables** in preparation for predictive modeling.
8. **Communicate insights and data-quality decisions** clearly and reproducibly.

Part 1: Meet the Data

Import the class survey data into Python.

Before changing anything, investigate the dataset.

Answer the following:

- How many observations are in the dataset?
- How many variables are there?
- What does one row represent?
- What does each column represent?
- Which variables appear to be numerical?
- Which variables appear to be categorical?
- Which variables contain free-form text?
- Are there variables whose current data type does not match what the variable actually represents?
- Are there missing values?

Create a small **data dictionary** containing, at minimum:

Variable	Description	Original Type	Desired Type
...	...	...	...

Part 2: Clean the Data

Carefully inspect **every variable** in the dataset.

Your goal is to create an analysis-ready version of the class data.

At minimum, consider the following.

Column Names

Rename columns so that they are concise, meaningful, and convenient to use in code.

For example, a survey question such as

How many hours of sleep do you typically get each night?

might become

`sleep_hours`

Use a consistent naming convention.

Data Types

Convert variables to appropriate data types.

Examples include:

- integer,
- float,
- categorical,
- Boolean,

- datetime,
- string/text.

Remember that a variable containing numbers is **not necessarily a numerical variable**.

Categorical Variables

Inspect the unique values of categorical variables.

Where appropriate, recode entries so that equivalent responses are represented consistently.

For example,

Python, python, PYTHON

should probably not represent three different categories.

Be thoughtful when recoding. **Do not combine categories simply because they look similar unless they represent the same information.**

Numerical Variables

Inspect numerical variables for:

- impossible values,
- unusual values,
- incorrect data types,
- inconsistent units, and
- potential outliers.

Do not automatically remove unusual observations. Determine whether they appear to be errors or legitimate values.

Missing Data

Identify missing values and determine how they are represented.

Do not automatically drop observations containing missing data.

For this lab, your cleaned dataset should preserve missing values unless you have a clear reason for changing them.

Document Your Decisions

Create a brief table summarizing the most important transformations you made.

Variable	Issue	Transformation
...	...	...

Part 3: Summarize the Data

Now describe your classmates.

Numerical Variables

For each meaningful numerical variable, calculate appropriate summaries such as:

- count,
- mean,
- standard deviation,
- median,
- minimum,
- maximum,
- quartiles.

Do not simply print a table and move on. Identify anything interesting or unexpected.

Categorical Variables

For categorical variables, examine:

- number of unique categories,
- frequency counts,
- proportions,
- most common responses.

For variables with many categories or free-form responses, determine an appropriate way to summarize them.

Part 4: Univariate Exploration

Create visualizations that show the distribution of individual variables.

Choose the visualization based on the type of variable rather than using the same graph for everything.

Possible choices include:

- histogram,
- box plot,
- bar chart,
- count plot,
- density plot,
- word cloud.

You do **not** need to use every type of plot listed above.

Instead, choose plots that help you understand the data.

For each visualization, include:

- a descriptive title,
- clearly labeled axes,
- appropriate units where applicable.

Briefly describe anything interesting you observe.

Part 5: Bivariate Exploration

Now investigate relationships between variables.

Select **at least three interesting pairs of variables** from the dataset.

Try to include different variable-type combinations when possible.

Numerical $\times$ Numerical

Examples of useful visualizations include:

- scatter plots,
- scatter plots with trend lines,
- correlation plots.

Categorical $\times$ Numerical

Examples include:

- side-by-side box plots,
- violin plots,
- grouped summary plots.

Categorical $\times$ Categorical

Examples include:

- grouped bar charts,
- stacked bar charts,
- heatmaps of counts or proportions.

For each relationship, briefly explain:

1. Why you chose the variables.
2. What the visualization shows.
3. Whether you see evidence of an interesting relationship.

Remember: Association does not imply causation.

Part 6: Multivariate Exploration

Real predictive models typically use information from multiple variables simultaneously.

Create **at least one visualization involving three or more variables**.

For example, you might represent additional variables using:

- color,
- marker shape,
- facets,
- grouped plots,
- pair plots.

Describe what the additional variable reveals that was not apparent in your bivariate analysis.

Part 7: Think Like a Predictive Modeler

We have not built a predictive model yet—but we can start thinking about one.

Choose **one variable in the class dataset that might serve as an interesting outcome variable**.

Suppose

Y = your chosen outcome.

Answer the following:

1. What would you like to predict?
2. Is this primarily a **regression** or **classification** problem?
3. Identify **3–5 variables** that might be useful predictors.
4. Explain briefly why you think each predictor could contain useful information about the outcome.
5. Are there variables you would **not** use as predictors? Why?
6. What data-quality or preprocessing issues would need to be addressed before fitting a model?

You are **not expected to fit a model** in this lab.

The goal is to begin connecting

Raw Data → Clean Data → Exploration → Prediction